

Figure 2: RealToxicityPrompts dataset experiment. **Left:** Empirical coverage rate (target 90%). The true coverage of the `Optimized` method is in a solid red line, while the upper and lower bounds on the coverage of the `Uncalibrated` and `Naive` methods correspond to the dotted lines. **Right:** Mean LPB. Shaded regions represent the standard deviation, computed over 5 runs. Higher is better.

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Checklist

1. For all models and algorithms presented, check if you include:

   (a) A clear description of the mathematical setting, assumptions, algorithm, and/or model. [Yes, all of our theory is given in Appendix D. Our algorithms are provided in Appendix B.]